Chapter 1

Propositional Logic

1.1 Proposition

Propositions are the basic building blocks of propositional logic.

Definition : A ***proposition*** is a sentence which is either true or false but not both.

The following are examples of propositions:

1. The capital of Canada is Toronto.
2. 1 plus 0 is 10.
3. The earth is revolving around the sun.
4. Every prime number greater than 2 is an odd integer.
5. Some universities do not have a computer science program.

Sentence 1 is false, because the capital of Canada is Ottawa. Sentence 2 is obviously false. Sentence 3 is true even though it was thought to be false five hundred years ago. Sentence 4 is true because every even integer greater than 2 is divisible by 2 and is thus not a prime number. Sentence 5 is true. One can identify one such university by browsing through the home pages of universities on the World Wide Web.

The following are some sentences whose status in propositions are not as easy to verify.

1. There are inhabitants on Mars.
2. Tiggy attends school at the First Lutheran Christian Academy.
3. Paul is a fan of the Detroit Red Wings.

Sentence 1 has to be either true or false but not both, even though we are not sure which one it is at the present time. Each of sentences 2 and 3 is also either true or false and not both, once we have identified the specific person we are talking about in the sentence.

Obviously, not all sentences are propositions. The following are some examples.

1. Shut the door!
2. Do you know who took the books from my desk?
3. Timothy is tall.
4. Good morning, Mr. Gray.

The first sentence is an imperative sentence which is neither true nor false. The second sentence is an interrogative sentence which is also neither true nor false. The third sentence could be both true and false. This is because it is quite likely that for people who are shorter than Timothy, Timothy is considered tall while for people who are taller than him, he is considered short. The last sentence is a greeting which is neither true nor false.

The main interest in propositional logic is not in how to identify propositions but in the operators that are applicable to propositions. We shall thus use symbols, such as p, q, r, ..., etc., to denote propositions. Each of these symbols is called a ***propositional variable***. This situation is analogous to that in high school algebra, where variables are used to denote numbers. However, in contrast with algebra in which a variable can be assigned any real number, there are only two values which can be assigned to any propositional variable. They are the ***truth value***s: *true* and *false*. We say that a propositional variable is true if it is assigned the *true* value and is false if it is assigned the *false* value. We shall use the abbrevations T and F to represent *true* and *false*, respectively.

1.2 Logical Operators

A ***logical operator*** is an operator which takes in logical values as operands and produces a logical value as the result. Five most commonly used logical operators are studied in this course. They are the *not*, *and*, *or*, *implication*, and *biconditional* operators.

The *not* operator has only one operand and is called a ***unary operator***. The remaining four logical operators each have two operands and are called ***binary operator***s. Since each operand of the logical operators can assume only one of the two values: *true* or *false*, there are altogether two different assignments of values to the operand of an unary operator, and four different assignments of values to the operands of a binary operator. As a result, the definitions of the logical operators can be given in tabular form as shown in Figure 1.2.1.

p	q	$p \diamond q$
F	F	v_1
F	T	v_2
T	F	v_3
T	T	v_4

(a) binary logical operators

p	$\diamond p$
F	v_1
T	v_2

(b) unary logical operators

Figure 1.2.1 Definitions of logical operators in tabular form. The v_i's are logical values

1. The *not* operator

The ***not*** operator, also called the ***negation*** operator is a unary operator. It is usually denoted by the symbol $\sim$. The following is its definition in tabular form.

p	$\sim p$
F	T
T	F

From the above definition, if p is false, then $\sim p$ is true, and vice versa. This is consistent with the usual meaning of *negation*. To see this, consider the proposition: "Detroit is to the north of Windsor." The negation of this proposition is the proposition: "Detroit is *not* to the north of Windsor." If the former is false, then the latter is true, and vice versa.

2. The *and* operator

The ***and*** operator, also called the ***conjunction*** operator is a binary operator. It is denoted by the symbol $\wedge$ and is defined as follows.

p	q	$p \wedge q$
F	F	F
F	T	F
T	F	F
T	T	T

We shall read $p \wedge q$ as "*p and q*." From the above definition, $p \wedge q$ is false whenever at least one of p and q is false, and is true otherwise. This definition of the logical operator "*and*" is consistent with the usual meaning of "*and*" in our daily conversation, as is illustrated in the following example.

Suppose I tell my son Pedro: "I will take you out to watch '*Twin Warriors*' *and* I will buy you a snack, tonight." Surely, Pedro will say that I have broken my promise (i.e., I have made a false statement) if either I fail to take him out to watch '*Twin Warriors*' or I fail to buy him a snack. He will happily say that I have kept my promise only if I have done both.

3. The *or* operator

The ***or*** operator, also called the ***disjunction*** operator is a binary operator. It is denoted by the symbol $\vee$ and is defined as follows.

p	q	$p \vee q$
F	F	F
F	T	T
T	F	T
T	T	T

We shall read $p \vee q$ as "*p or q*." From the above definition, $p \vee q$ is true whenever at least one of p and q is true, and is false otherwise.

Consider the statement: "I will take you out to watch '*Twin Warriors*' *or* I will buy you a snack tonight." This statement consists of two constituent propositions: "I will take you out to watch '*Twin Warriors*,' tonight" and "I will buy you snack, tonight." It is true if at least one of the two constituent propositions is true, and is false otherwise. This usage of 'or' in daily conversation is consistent with the definition of $\vee$.

Now, consider the following statement: "Peter published the paper in SIAM *Journal on Computing* or in J.ACM." This statement consists of two constituent propositions: "Peter published the paper in

SIAM *Journal on Computing*" and "Peter published the paper in J.ACM." Since publication of the same paper in different journals is disallowed, the two constituent propositions cannot both be true. In other words, the statement is false if both of its constituent propositions are true. This usage of 'or' in daily conversation is inconsistent with the definition of $\vee$.

The conjunctive 'or' is said to be used in the inclusive sense in the first statement and in the exclusive sense in the second statement. Hence, $\vee$ is also called the ***inclusive-or*** operator. The logical operator corresponding to the exclusive sense usage of 'or' is called the ***exclusive-or*** operator and is normally denoted by $\oplus$. The exclusive-or operator is not studied in these notes.

4. The *conditional* operator

The ***conditional*** operator, also called the ***implication*** operator is a binary operator. It is denoted by the symbol $\Rightarrow$ and is defined as follows.

p	q	$p \Rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

In $p \Rightarrow q$, p is called the ***premise, hypothesis***, or ***antecedent***; q is called the ***conclusion*** or ***consequent***, and $p \Rightarrow q$ is read "*if p then q*."

Let us examine the above definition. When p is T and q is T, $p \Rightarrow q$ is T, and when p is T and q is F, $p \Rightarrow q$ is F. The intended meaning of $p \Rightarrow q$ in these two cases is that the truth of p implies that of q which is consistent with the usual meaning of "*if p then q*." As an example, consider the following statement.

"If you give me \$10, I will give you a copy of these notes."

Suppose you give me \$10, and I give you a copy of these notes. Then, the statement is regarded as true in our daily conversation. On the other hand, if you give me \$10, but I refuse to give you a copy of these notes, then the statement is regarded as false in our daily conversation.

For the two remaining cases (i.e., the two cases in which p is F), we observe that $p \Rightarrow q$ is T, regardless of the value of q. The reason why the value of $p \Rightarrow q$ is so assigned can be explained as follows. "If p then q" only asserts that if p is true, then q is true. It makes no assertion as to what q is if p is false. So, if p is false, "If p then q" should *not* be regarded as false, regardless of the value of q. In other words, if p is F, it is plausible to assert that $p \Rightarrow q$ is T, regardless of the value of q. As an example, consider again the statement: "If you give me \$10, I will give you a copy of these notes." If you do not give me \$10, then you will have no right in accusing me of having made a false statement, regardless of whether I give you a copy of these notes or not.

Most of the theorems in mathematics are of the form $p \Rightarrow q$. It is for this reason that the conditional operator is of special interest to us. Besides being read as "if p then q," $p \Rightarrow q$ may also be read in one of the following ways:

p implies q
p only if q
q if p
p is a sufficient condition for q

q is a necessary condition for p
q provided that p
q is a consequence of p
q whenever p
not p unless q
not p or q

Note that the last two ways of reading out $p \Rightarrow q$ actually states the conditions, namely q is T or p is F, under which $p \Rightarrow q$ is T.

5. The *biconditional* operator

The ***biconditional*** operator , also called the ***if and only if*** operator is a binary operator. It is denoted by the symbol $\Leftrightarrow$ and is defined as follows.

p	q	$p \Leftrightarrow q$
F	F	T
F	T	F
T	F	F
T	T	T

We shall read $p \Leftrightarrow q$ in one of the following ways:

p if and only if q
p is a sufficient and necessary condition for q

p if and only if q is usually abbreviated as *p iff q*.

From the above definition, $p \Leftrightarrow q$ is true whenever p and q are both false, or both true. i.e. they have the same value. It is false, otherwise. This is consistent with the usual meaning of 'if and only if' in daily conversation. The following is an illustration.

Consider the statement: "I give you a copy of these notes *if and only if* you give me $10." The statement is true if either I give you a copy of these notes and you give me $10, or I do not give you a copy of these notes and you do not give me $10. It is false if either I give you a copy of these notes and you do not give me $10, or I do not give you a copy of these notes and you give me $10.

1.3 Well-formed Formulas

As with algebra in which arithmetic operators are used to chain variables together to form algebraic expressions, logical operators can be used to chain propositional variables together to form logical expressions. Logical expressions are called ***well-formed formula***s (abbreviated as ***wff***s) in propositional logic.

Definition:

(a) All propositional variables and the constants T(*true*) and F(*false*) are wffs.
(b) If α and β are wffs, then so are $\sim\alpha$, $(\alpha \wedge \beta)$, $(\alpha \vee \beta)$, $(\alpha \Rightarrow \beta)$ and $(\alpha \Leftrightarrow \beta)$.
(c) No expression other than those which can be defined by means of a finite number of applications of (a), (b) are wffs.

In the following example, we demonstrate how to use the above definition to generate wffs.

Example 1.3.1 Let p, q be two propositional variables. By rule (a), p and q are wffs.

Starting with p and q, a number of wffs can be generated as follows.

Let p be α, q be β. By rule (b), $(p \vee q)$ is a wff.

Since $(p \vee q)$ is a wff, by letting $(p \vee q)$ be α, and by rule (b) again, $\sim (p \vee q)$ is a wff. Since p and $\sim (p \vee q)$ are wffs, by letting p be α, $\sim (p \vee q)$ be β, and using rule (b), $(p \Rightarrow \sim (p \vee q))$ is a wff.

Obviously, this process can continue as long as we like. ■

From the definition of wff and the above example, it is easily seen that a wff may contain other wffs. For instance, the wff $\sim (p \vee q)$ contains the wffs: p, q, $(p \vee q)$, and itself. The last wff in Example 1.3.1 contains every wff (including itself) generated in that example. This motivates us to define the following.

Definition: A ***sub well-formed formula*** (abbrivated as ***sub-wff***) of a wff is a wff it contains.

Therefore, p, q, $(p \vee q)$ and $\sim (p \vee q)$ are sub-wff of $\sim (p \vee q)$. Similarly, every wff generated in the Example 1.3.1 is a sub-wff of $(p \Rightarrow \sim (p \vee q))$.

Using the definition of wff, we can check if a given sequence of symbols forms a wff. The idea is illustrated in the following two examples.

Example 1.3.2 Show that the following are wffs.

(i) $(p \Rightarrow q)$

(ii) $(p \Rightarrow ((r \vee \sim q) \vee r))$

(iii) $\sim (p \wedge (q \Rightarrow r))$

(iv) $((p \vee \sim (p \Rightarrow q)) \Rightarrow (p \wedge \sim (q \vee (p \wedge q))))$

(v) $(((p \Leftrightarrow q) \wedge (r \Leftrightarrow s)) \vee (t \Leftrightarrow s))$

Solution: We need only show that each of the above sequences of symbols can be constructed by applying rules (a) and (b) in the definition of wff a finite number of times. For then, by rule (c), the sequence of symbols forms a wff. For clarity, we shall present our solution in the following format: all the wffs generated by the rules in the course of constructing the given wff are listed vertically in the order in which they are generated. To the right of each wff in the list, the rule ((a) or (b)) used in generating the wff and how it is used are specified. Moreover, in applying rule (b), if ω is a wff which replaces, say α, then instead of writing 'let ω be α', we shall write 'α=ω.'

(i) $(p \Rightarrow q)$ can be constructed by two applications of rule (a) and one application of rule (b) as follows. So by rule (c), it is a wff.

p	rule (a), p is a propositional variable
q	rule (a), q is a propositional variable
$(p \Rightarrow q)$	rule (b), $(\alpha \Rightarrow \beta)$ with $\alpha = p$, $\beta = q$

(ii) $(p \Rightarrow ((r \vee \sim q) \vee r))$ can be constructed by three applications of rule (a) and four applications of rule (b) as follows. Hence, by rule (c), it is a wff.

q	rule (a), q is a propositional variable
$\sim q$	rule (b), $\sim \alpha$ with $\alpha = q$
r	rule (a), r is a propositional variable
$(r \vee \sim q)$	rule (b), $(\alpha \vee \beta)$, with $\alpha = r$, $\beta = \sim q$
$((r \vee \sim q) \vee r)$	rule (b), $(\alpha \vee \beta)$, with $\alpha = (r \vee \sim q)$, $\beta = r$
p	rule (a), p is a propositional variable
$(p \Rightarrow ((r \vee \sim q) \vee r))$	rule (b), $(\alpha \Rightarrow \beta)$ with $\alpha = p$, $\beta = ((r \vee \sim q) \vee r)$

(iii) $\sim (p \wedge (q \Rightarrow r))$ can be constructed by three applications of rule (a) and three applications of rule (b) as follows. Hence by rule (c), it is a wff.

q	rule (a), q is a propositional variable
r	rule (a), r is a propositional variable
$(q \Rightarrow r)$	rule (b), $(\alpha \Rightarrow \beta)$ with $\alpha = q, \beta = r$
p	rule (a), p is a propositional variable
$(p \wedge (q \Rightarrow r))$	rule (b), $(\alpha \wedge \beta)$, with $\alpha = p, \beta = (q \Rightarrow r)$
$\sim (p \wedge (q \Rightarrow r))$	rule (b), $\sim \alpha$ with $\alpha = (p \wedge (q \Rightarrow r))$

(iv) $((p \vee \sim (p \Rightarrow q)) \Rightarrow (p \wedge \sim (q \vee (p \wedge q))))$ can be constructed by two applications of rule (a) and eight applications of rule (b) as follows. Hence, by rule (c), it is a wff.

p	rule (a), p is a propositional variable
q	rule (a), q is a propositional variable
$(p \Rightarrow q)$	rule (b), $(\alpha \Rightarrow \beta)$ with $\alpha = p, \beta = q$
$\sim (p \Rightarrow q)$	rule (b), $\sim \alpha$ with $\alpha = (p \Rightarrow q)$
$(p \vee \sim (p \Rightarrow q))$	rule (b), $(\alpha \vee \beta)$ with $\alpha = p, \beta = \sim (p \Rightarrow q)$
$(p \wedge q)$	rule (b), $(\alpha \wedge \beta)$ with $\alpha = p, \beta = q$
$(q \vee (p \wedge q))$	rule (b), $(\alpha \vee \beta)$ with $\alpha = q, \beta = (p \wedge q)$
$\sim (q \vee (p \wedge q))$	rule (b), $\sim \alpha$ with $\alpha = (q \vee (p \wedge q))$
$(p \wedge \sim (q \vee (p \wedge q)))$	rule (b), $(\alpha \wedge \beta)$ with $\alpha = p, \beta = \sim (q \vee (p \wedge q))$
$((p \vee \sim (p \Rightarrow q)) \Rightarrow (p \wedge \sim (q \vee (p \wedge q))))$	rule (b), $(\alpha \Rightarrow \beta)$ with $\alpha = (p \vee \sim (p \Rightarrow q))$, $\beta = (p \wedge \sim (q \vee (p \wedge q)))$

(v) $(((p \Leftrightarrow q) \wedge (r \Leftrightarrow s)) \vee (t \Leftrightarrow s))$ can be constructed by five applications of rule (a) and five applications of rule (b) as follows. Hence by rule (c), it is a wff.

p	rule (a), p is a propositional variable
q	rule (a), q is a propositional variable
$(p \Leftrightarrow q)$	rule (b), with $(\alpha \Leftrightarrow \beta)$, where $\alpha = p, \beta = q$
r	rule (a), r is a propositional variable
s	rule (a), s is a propositional variable
$(r \Leftrightarrow s)$	rule (b), with $(\alpha \Leftrightarrow \beta)$, where $\alpha = r, \beta = s$
t	rule (a), t is a propositional variable
$(t \Leftrightarrow s)$	rule (b), with $(\alpha \Leftrightarrow \beta)$, where $\alpha = t, \beta = s$
$((p \Leftrightarrow q) \wedge (r \Leftrightarrow s))$	rule (b), $(\alpha \wedge \beta)$ with $\alpha = (p \Leftrightarrow q), \beta = (r \Leftrightarrow s)$
$(((p \Leftrightarrow q) \wedge (r \Leftrightarrow s)) \vee (t \Leftrightarrow s))$	rule (b), $(\alpha \vee \beta)$ with $\alpha = ((p \Leftrightarrow q) \wedge (r \Leftrightarrow s))$, $\beta = (t \Leftrightarrow s)$ ■

Example 1.3.3 Show that the following are *not* wffs.

(i) $\sim q \vee p \Leftrightarrow q$
(ii) $\sim (p)$
(iii) $((p \wedge p \vee q) \vee r)$
(iv) $((p \wedge q) \vee (\sim (r \Rightarrow q)) \wedge \sim (p \vee q))$

Solution: We shall show that each of the sequences of symbols cannot be constructed using the rules (a) and (b) in the definition of wff a finite number of times. Then, by rule (c), the sequence is not a wff.

(i) From the definition of wff, we notice that corresponding to every binary logical operator, there is a distinct pair of parentheses. Since $\sim q \vee p \Leftrightarrow q$ contains two binary operators and no parentheses, it cannot be a wff.

(ii) By the definition of wff, neither the propostional variables nor the unary operator $\sim$ comes with parentheses, and there is no other way to introduce parentheses without involving binary operator. So, it is impossible to construct $\sim (p)$ using rules (a) and (b). By rule (c), $\sim (p)$ is not a wff.

(iii) As was pointed out in Part 1, there is a one-to-one correspondence between the binary operators and the pairs of parenthese in every wff. Since there are three binary operators and two pairs of parenthese in $((p \wedge p \vee q) \vee r)$, it cannot be a wff.

(iv) We shall show that the given sequence of symbols cannot be constructed by using rules (a) and (b). First, note that we must begin the construction from the propositional variables. So, we apply rule (a) to claim that each of p, q and r is a wff. To include the first $\wedge$ operator, the only way is to let α be p and β be q, and then apply rule (b). This results in the sub-wff $(p \wedge q)$. A similar argument applies to the $\Rightarrow$ operator and the second $\vee$ operator which results in the sub-wffs, $(r \Rightarrow q)$ and $(p \vee q)$, respectively. Next, to include the first $\sim$ operator, the only way is to let α be $(r \Rightarrow q)$ and then apply rule (b). This results in $\sim (r \Rightarrow q)$. In a similar way, we obtain the sub-wff $\sim (p \vee q)$ from $(p \vee q)$. Now, it remains to include the remaining $\vee$ and $\wedge$ operators. If we include $\vee$ before $\wedge$, we will obtain the wff $(((p \wedge q) \vee \sim (r \Rightarrow q)) \wedge \sim (p \vee q))$ which is not identical to the given sequence of symbols. On the other hand, if we include $\wedge$ before $\vee$, we will obtain the wff $((p \wedge q) \vee (\sim (r \Rightarrow q) \wedge \sim (p \vee q)))$ which again is not identical to the given sequence of symbols. Therefore the given sequence of symbols cannot be constructed by applying rules (a) and (b) a finite number of times. It is thus not a wff by rule (c). ■

In the above example, we already observed that corresponding to each binary operator, there is a distinct pair of parentheses. Therefore, in a wff with n binary operators, there are $2n$ parentheses. When n is large (five or more), the parentheses in the wff could be a nuisance. For convenience, we shall try to reduce the number of parentheses involved. This can be achieved by using the following operator precedence and left associativity rule.

Precedence of logical operators:

Operator	Precedence
$\sim$	1
$\wedge$	2
$\vee$	3
$\Rightarrow$	4
$\Leftrightarrow$	5

Associativity of logical operators: In the absence of parentheses,

the binary operators are ***left-associative***. i.e., between every two consecutive binary operators of the same type, the one on the left has higher precedence than the one on the right.

Remark: From rule (b), it is obvious that the unary operator $\sim$ is ***right-associative***. i.e., between every two consecutive $\sim$ operators, the one on the right has higher precedence than the one on the left.

An operator with precedence higher than its neighbouring operator(s) binds more tightly to its operand(s) than the neighbouring operator(s) and the parentheses surrounding it can be dropped. This gives rise to the following set of rules for abbreviating wffs.

(i) Let $\diamond$ and $\circ$ be binary logical operators such that $\diamond$ has higher precedence than $\circ$. The inner pair of parenthese of any sub-wff of the form $((\alpha \diamond \beta) \circ \gamma)$ or $(\alpha \circ (\beta \diamond \gamma))$ can be removed, where α, β, γ are wffs.

(ii) Let $\odot$ be any binary logical operator. Any sub-wff of the form $((\alpha \odot \beta) \odot \gamma)$ can be simplified to $(\alpha \odot \beta \odot \gamma)$, where α, β, γ are wffs.

(iii) Obviously, the outermost pair of parentheses (if exists) can always be removed without causing any confusion.

The reader may have noticed that none of the rules applies to the unary operator $\sim$. This is because each inclusion of $\sim$ to a wff does not introduce any parentheses.

Example 1.3.4 Eliminate as many parentheses as possible from each of the following wffs.

(i) $(p \Rightarrow ((r \vee \sim q) \vee r))$

(ii) $\sim (p \wedge (q \Rightarrow r))$

(iii) $((p \vee \sim (p \Rightarrow q)) \Rightarrow (p \wedge \sim (q \vee (p \wedge q))))$

(iv) $(((p \Leftrightarrow q) \wedge (r \Leftrightarrow s)) \vee (t \Leftrightarrow s))$

Solution:

(i) By rule (ii), the inner pair of parenthese of the sub-wff $((r \vee \sim q) \vee r)$ can be removed. In the sub-wff $(p \Rightarrow ((r \vee \sim q) \vee r))$, by letting the $\Rightarrow$ operator be $\circ$ and the second $\vee$ operator be $\diamond$, the pair of parentheses enclosing the sub-wff $(r \vee \sim q) \vee r$ can be removed. Finally, by rule (iii), the outermost parenthese can be removed. The resulting abbreviated wff is $p \Rightarrow r \vee \sim q \vee r$.

(ii) No rule can be applied to $\sim (p \wedge (q \Rightarrow r))$. Therefore, the wff cannot be abbreviated.

(iii) By rule (i), the inner pair of parenthese of the sub-wffs $(q \vee (p \wedge q))$ can be removed. No rule can be applied to the sub-wffs $(p \wedge \sim (q \vee (p \wedge q)))$ and $(p \vee \sim (p \Rightarrow q))$. As for the sub-wff $((p \vee \sim (p \Rightarrow q)) \Rightarrow (p \wedge \sim (q \vee (p \wedge q))))$. If we let p be α, $\sim (p \Rightarrow q)$ be β and $(p \wedge \sim (q \vee (p \wedge q)))$ be γ, we would obtain a sub-wff of the form $((\alpha \vee \beta) \Rightarrow \gamma)$. Since $\vee$ has higher precedence than $\Rightarrow$, by rule (i), the parentheses enclosing $p \vee \sim (p \Rightarrow q)$ can be removed. On the other hand, if we let $(p \vee \sim (p \Rightarrow q))$ be α, p be β and $\sim (q \vee (p \wedge q))$ be γ, we would obtain a sub-wff of the form $(\alpha \Rightarrow (\beta \wedge \gamma))$. As $\wedge$ has higher precedence than $\Rightarrow$, by rule (i), the parentheses enclosing $p \wedge \sim (q \vee (p \wedge q))$ can be removed. Finally, by rule (iii), the outermost pair of parenthese can be removed. This results in $p \vee \sim (p \Rightarrow q) \Rightarrow p \wedge \sim (q \vee p \wedge q)$.

(iv) Since $\wedge$ has higher precedence than $\Leftrightarrow$, none of the parenthese in $(p \Leftrightarrow q)$ and $(r \Leftrightarrow s)$ can be eliminated. On the other hand, as $\wedge$ has higher precedence than $\vee$, by rule (i), the pair of parentheses enclosing $(p \Leftrightarrow q) \wedge (r \Leftrightarrow s)$ can be eliminated. This results in $((p \Leftrightarrow q) \wedge (r \Leftrightarrow s) \vee (t \Leftrightarrow s))$. Finally, by rule (iii), the outermost pair of parentheses can be removed. The result is $(p \Leftrightarrow q) \wedge (r \Leftrightarrow s) \vee (t \Leftrightarrow s)$. ■

The above method of abbreviating wffs would be invalid unless we can restore the parentheses of any abbreviated wff unambiguously. The following is a set of rules for this purpose.

(i') In each of the following form of abbreviated sub-wffs, a pair of parentheses can be introduced to enclose the term $\beta \diamond \gamma$, provided that the operator $\diamond$ has higher precedence than that of $\otimes$ and is not lower than that of $\odot$.

$(a) \cdots \alpha \otimes \beta \diamond \gamma \odot \delta \cdots$, $(b) \cdots \alpha \otimes \beta \diamond \gamma$, $(c) (\cdots \alpha \otimes \beta \diamond \gamma)$, $(d) \beta \diamond \gamma \odot \delta \cdots$ and $(e) (\beta \diamond \gamma \odot \delta \cdots)$.

(ii') If the given abbreviated wff is not of the form $\sim \alpha$ where α is an abbreviated wff, then a pair of parentheses can be introduced to enclose the given abbreviated wff.

Obviously, rule (i') undoes the work of rules (i) and (ii) while rule (ii') undoes the work of rule (iii).

Example 1.3.5 Restore all the parentheses in the following abbreviated wffs:

(i) $r \vee \sim q \vee r \wedge p$

(ii) $\sim w \vee (p \Rightarrow \sim q \Rightarrow r) \vee p \wedge s \wedge t$

(iii) $\sim (p \Leftrightarrow \sim (p \Rightarrow q) \Leftrightarrow p \wedge \sim (q \vee p \wedge q))$

Solution: We shall present our solutions in a step-wise manner in which every step restores one pair of parentheses using one of the rules (i') and (ii').

(i) Given $r \vee \sim q \vee r \wedge p$.

$\left(\underbrace{r}_{\beta} \vee \underbrace{\sim q}_{\gamma}\right) \vee \underbrace{r}_{\delta} \wedge p$ Rule(i'), using form : $\beta \diamond \gamma \odot \delta \cdots$

$\underbrace{(r \vee \sim q)}_{\alpha} \vee \left(\underbrace{r}_{\beta} \wedge \underbrace{p}_{\gamma}\right)$ Rule(i'), using form : $\alpha \otimes \beta \diamond \gamma$

$((r \vee \sim q) \vee (r \wedge p))$ Rule(ii')

(ii) Given $\sim w \vee (p \Rightarrow \sim q \Rightarrow r) \vee p \wedge s \wedge t$.

$$\Big(\underbrace{\sim w}_{\beta} \vee \underbrace{(p \Rightarrow \sim q \Rightarrow r)}_{\gamma}\Big) \vee \underbrace{p}_{\delta} \wedge s \wedge t \qquad \text{Rule(i}'\text{), using form : } \beta \diamond \gamma \odot \delta \cdots$$

$$\underbrace{(\sim w \vee (p \Rightarrow \sim q \Rightarrow r))}_{\alpha} \vee \Big(\underbrace{p}_{\beta} \wedge \underbrace{s}_{\gamma}\Big) \wedge \underbrace{t}_{\delta} \qquad \text{Rule(i}'\text{), using form : } \alpha \otimes \beta \diamond \gamma \odot \delta$$

$$\underbrace{(\sim w \vee (p \Rightarrow \sim q \Rightarrow r))}_{\alpha} \vee \Big(\underbrace{(p \wedge s)}_{\beta} \wedge \underbrace{t}_{\gamma}\Big) \qquad \text{Rule(i}'\text{), using form : } \cdots \alpha \otimes \beta \diamond \gamma$$

$$\Big(\sim w \vee \Big(\Big(\underbrace{p}_{\beta} \Rightarrow \underbrace{\sim q}_{\gamma}\Big) \Rightarrow \underbrace{r}_{\delta}\Big)\Big) \vee ((p \wedge s) \wedge t) \qquad \text{Rule(i}'\text{), using form : } (\beta \diamond \gamma \odot \delta \cdots)$$

$$((\sim w \vee ((p \Rightarrow \sim q) \Rightarrow r)) \vee ((p \wedge s) \wedge t)) \qquad \text{Rule(ii}'\text{)}$$

(iii) Given $\sim (p \Leftrightarrow \sim (p \Rightarrow q) \Leftrightarrow p \wedge \sim (q \vee p \wedge q))$.

$$\sim \Big(\Big(\underbrace{p}_{\beta} \Leftrightarrow \underbrace{\sim (p \Rightarrow q)}_{\gamma}\Big) \Leftrightarrow \underbrace{p}_{\delta} \wedge \sim (q \vee p \wedge q)\Big) \qquad \text{rule(i}'\text{), using form : } (\beta \diamond \gamma \odot \delta \cdots)$$

$$\sim \Big((p \Leftrightarrow \sim (p \Rightarrow q)) \Leftrightarrow p \wedge \sim \Big(\underbrace{q}_{\alpha} \vee \Big(\underbrace{p}_{\beta} \wedge \underbrace{q}_{\gamma}\Big)\Big)\Big) \qquad \text{rule(i}'\text{), using form : } (\cdots \alpha \otimes \beta \diamond \gamma)$$

$$\sim \Big(\underbrace{(p \Leftrightarrow \sim (p \Rightarrow q))}_{\alpha} \Leftrightarrow \Big(\underbrace{p}_{\beta} \wedge \underbrace{\sim (q \vee (p \wedge q))}_{\gamma}\Big)\Big) \qquad \text{rule(i}'\text{), using form : } (\alpha \otimes \beta \diamond \gamma)$$

Note that rule (ii') cannot be applied in this example. ■

In the remaining part of this notes, the term *wff* (well-formed formula) refers to both abbreviated and unabbreviated wffs.

1.4 The Truth Table

In high school algebra, we frequently encountered situations in which we were given an algebraic expression and an assignment of values to the variables in the expression, and were told to determine the value of that expression. The process of determining the value of an algebraic expression is called an *evaluation* of the expression.

Since wffs are logical expressions, they can be evaluated in the above sense. However, in evaluating a wff, only truth values can be assigned to the variables in the wff. Moreover, the value of the wff is determined by virtue of the definitions of the logical operators, the priority scheme of the operators, and the grouping rule.

The process of evaluating a wff is similar to that of an algebraic expression. The following is an illustration.

Example 1.4.1 Evaluate the wff : $p \vee \sim (p \Rightarrow q) \Rightarrow p \wedge \sim (q \vee p \wedge q)$ using the following assignment of truth values to its variables: $p = \mathsf{T}$, $q = \mathsf{F}$.

Solution: First, restore all the parentheses to obtain the fully parenthesized wff: $((p \vee \sim (p \Rightarrow q)) \Rightarrow (p \wedge \sim (q \vee (p \wedge q))))$.

Next, substitute T for p and F for q in the fully parenthesized wff to obtain: $((\mathrm{T} \vee \sim (\mathrm{T} \Rightarrow \mathrm{F})) \Rightarrow (\mathrm{T} \wedge \sim (\mathrm{F} \vee (\mathrm{T} \wedge \mathrm{F}))))$.

Then, choose an inner-most sub-wff (one whose operand(s) have been reduced to logical constant(s)) and replace it with a truth value by virtue of the definition of the operator it contains. Repeat this step until the given wff is reduced to a single truth value. That is the result of the evaluation.

The evaluation of the given wff is demonstrated below in a stepwise manner. In each step, the chosen innermost sub-wff is underlined.

$$\begin{array}{l}
\Big(\big(\mathrm{T} \vee \sim \underline{(\mathrm{T} \Rightarrow \mathrm{F})}\big) \Rightarrow (\mathrm{T} \wedge \sim (\mathrm{F} \vee (\mathrm{T} \wedge \mathrm{F})))\Big) \\
((\mathrm{T} \vee \underline{\sim \mathrm{F}}) \Rightarrow (\mathrm{T} \wedge \sim (\mathrm{F} \vee (\mathrm{T} \wedge \mathrm{F})))) \\
\Big(\underline{(\mathrm{T} \vee \mathrm{T})} \Rightarrow (\mathrm{T} \wedge \sim (\mathrm{F} \vee (\mathrm{T} \wedge \mathrm{F})))\Big) \\
\Big(\mathrm{T} \Rightarrow \big(\mathrm{T} \wedge \sim \big(\mathrm{F} \vee \underline{(\mathrm{T} \wedge \mathrm{F})}\big)\big)\Big) \\
\Big(\mathrm{T} \Rightarrow \big(\mathrm{T} \wedge \sim \underline{(\mathrm{F} \vee \mathrm{F})}\big)\Big) \\
(\mathrm{T} \Rightarrow (\mathrm{T} \wedge \underline{\sim \mathrm{F}})) \\
\Big(\mathrm{T} \Rightarrow \underline{(\mathrm{T} \wedge \mathrm{T})}\Big) \\
\underline{(\mathrm{T} \Rightarrow \mathrm{T})} \\
\mathrm{T}
\end{array}$$

Hence, $p \vee \sim (p \Rightarrow q) \Rightarrow p \wedge \sim (q \vee p \wedge q)$ is evaluated to T, when p is T, q is F. ■

Now, consider a wff with n propositional variables. Since there are 2 different ways of assigning a truth value to each of the variables, there are altogether 2^n different assignments of truth values to the n propositional variables. Consequently, the wff has altogether 2^n different evaluations. It is always convenient to compute the result of each of these evaluations and store the results for later reference. The *truth table* is a tool invented for this purpose.

Example 1.4.2 In the wff $\sim (p \Rightarrow q)$, each of the variables p and q can assume 2 different truth values. Therefore, there are altogether $2 \times 2 = 2^2 = 4$ different ways of assigning truth values to the two variables. For each of these assignments of truth values, the wff has a corresponding truth value. The following is the truth table for the wff $\sim (p \Rightarrow q)$.

p	q	$(p \Rightarrow q)$	$\sim (p \Rightarrow q)$
F	F	T	F
F	T	T	F
T	F	F	T
T	T	T	F

Note that the last column contains the values of $\sim (p \Rightarrow q)$ for all the assignments of truth values, and $(p \Rightarrow q)$ is a sub-wff of $\sim (p \Rightarrow q)$ whose values must be determined in order to determine that of $\sim (p \Rightarrow q)$. ■

In general, for every wff with n propositional variables, there are 2^n rows in the truth table. Each row exhibits an assignment of truth values to the propositional variables of the wff. For visual convenience, the 2^n assignments of truth values are normally ordered lexicographically with the provision that the truth value F precedes the truth value T. Moreover, for each sub-wff of the given wff, there corresponds a column and the truth values in that column are the values of the sub-wff corresponding to the 2^n different assignments of truth values.

Example 1.4.3 Construct a truth table for $(p \Leftrightarrow q) \wedge (r \Leftrightarrow s) \vee (p \Leftrightarrow s)$.
Solution: The sub-wffs in $(p \Leftrightarrow q) \wedge (r \Leftrightarrow s) \vee (p \Leftrightarrow s)$ are the following:

$p, q, r, s, (p \Leftrightarrow q), (r \Leftrightarrow s), (p \Leftrightarrow q) \wedge (r \Leftrightarrow s), (p \Leftrightarrow s), (p \Leftrightarrow q) \wedge (r \Leftrightarrow s) \vee (p \Leftrightarrow s)$

The truth table is as follow. ■

p	q	r	s	$(p \Leftrightarrow q)$	$(r \Leftrightarrow s)$	$(p \Leftrightarrow q) \wedge (r \Leftrightarrow s)$	$(p \Leftrightarrow s)$	$(p \Leftrightarrow q) \wedge (r \Leftrightarrow s) \vee (p \Leftrightarrow s)$
F	F	F	F	T	T	T	T	T
F	F	F	T	T	F	F	F	F
F	F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	F	T
F	T	F	F	F	T	F	T	T
F	T	F	T	F	F	F	F	F
F	T	T	F	F	F	F	T	T
F	T	T	T	F	T	F	F	F
T	F	F	F	F	T	F	F	F
T	F	F	T	F	F	F	T	T
T	F	T	F	F	F	F	F	F
T	F	T	T	F	T	F	T	T
T	T	F	F	T	T	T	F	T
T	T	F	T	T	F	F	T	T
T	T	T	F	T	F	F	F	F
T	T	T	T	T	T	T	T	T

1.5 Tautology, Contradiction and Satisfiable wff

Well-formed formulas which are always evaluated to *true* are of particular interest, for they represent statement forms which always hold true.

Definition: A ***tautology*** is a wff whose value is *true* for every possible assignment of truth values to its variables.

Definition: A ***contradiction*** is a wff whose value is *false* for every possible assignment of truth values to its variables.

Definition: A wff is ***satisfiable*** if its value is true for at least one assignment of truth values to its variables. A wff that is neither a tautology nor a contradiction is called a ***contingency***.

Since the truth table of a wff contains the value of every possible evaluation of that wff, it can be conveniently used as a means to determine if a wff is a tautology, a contradiction, a contingency, or is satisfiable, provided that the number of propositional variables involved is not too large.

Example 1.5.1 Prove that $((\sim p \vee q) \Leftrightarrow (p \Rightarrow q))$ is a tautology.

Solution: Construct the truth table of $((\sim p \vee q) \Leftrightarrow (p \Rightarrow q))$ as follows.

p	q	$\sim p$	$(\sim p \vee q)$	$(p \Rightarrow q)$	$((\sim p \vee q) \Leftrightarrow (p \Rightarrow q))$
F	F	T	T	T	T
F	T	T	T	T	T
T	F	F	F	F	T
T	T	F	T	T	T

Since the column for $((\sim p \vee q) \Leftrightarrow (p \Rightarrow q))$ in the truth table contains the T value only, the wff $((\sim p \vee q) \Leftrightarrow (p \Rightarrow q))$ is thus a tautology. ■

Example 1.5.2 Prove that $(((p \Rightarrow r) \wedge \sim r) \wedge p)$ is a contradiction.

Solution: Construct the truth table of $(((p \Rightarrow r) \wedge \sim r) \wedge p)$ as follows.

p	r	$\sim r$	$(p \Rightarrow r)$	$((p \Rightarrow r) \wedge \sim r)$	$(((p \Rightarrow r) \wedge \sim r) \wedge p)$
F	F	T	T	T	F
F	T	F	T	F	F
T	F	T	F	F	F
T	T	F	T	F	F

Since the column for $(((p \Rightarrow r) \wedge \sim r) \wedge p)$ in the truth table contains only the F value, the wff $(((p \Rightarrow r) \wedge \sim r) \wedge p)$ is thus a contradiction. ■

Example 1.5.3 Prove that $((p \wedge q) \Rightarrow \sim (p \vee q))$ is satisfiable.

Solution: Construct the truth table of $((p \wedge q) \Rightarrow \sim (p \vee q))$ as follows.

p	q	$(p \wedge q)$	$(p \vee q)$	$\sim (p \vee q)$	$(p \wedge q) \Rightarrow \sim (p \vee q)$
F	F	F	F	T	T
F	T	F	T	F	T
T	F	F	T	F	T
T	T	T	T	F	F

Since the column for $((p \wedge q) \Rightarrow \sim (p \vee q))$ in the truth table contains at least one instance of the T value, $((p \wedge q) \Rightarrow \sim (p \vee q))$ is thus satisfiable. In fact, it is also a contingency. ■

The following theorem states a relationship between tautology and contradiction.

Theorem 1.5.1: *A wff ω is a tautology if and only if $\sim \omega$ is a contradiction.*

Proof: ω is a tautology
iff ω is evaluated to true for every assignment of truth values to its variables
iff $\sim \omega$ is evaluated to false for every assignment of truth values to its variables
iff $\sim \omega$ is a contradiction. ■

1.6 Equivalences and Inference Rules

Definition: Two wffs α and β are ***logically equivalent***, denoted by $\alpha \equiv \beta$, if and only if $\alpha \Leftrightarrow \beta$ is a tautology.

Example 1.6.1 In Example 1.5.1, we showed that $(\sim p \vee q) \Leftrightarrow (p \Rightarrow q)$ is a tautology.
Therefore, $(\sim p \vee q) \equiv (p \Rightarrow q)$. ■

A list of equivalences is given in Table 1.6.1. The validity of each of them can be verified by using a *truth table*. However, before doing so, one must notice that while the operands α, β and γ in the equivalences are wffs, the truth table was originally introduced to evaluate wffs whose operands are propositional variables only. Fortunately, as every wff is evaluated to the value *true* or *false*, it is not difficult to see that the wffs can play the roles of the propositional variables, and the truth table can be easily generalized to evaluate wffs whose operands are not just propositional variables but wffs. As an example, consider the equivalence E16. To show that the equivalence holds, we must prove that $\sim (\alpha \wedge \beta) \Leftrightarrow (\sim \alpha \vee \sim \beta)$ is a tautology. Since the wff contains two sub-wffs, namely α and β, we can construct a truth table for two propositional variables, with α and β playing the role of the propositional variables. The truth table is as follows:

α	β	$\sim \alpha$	$\sim \beta$	$(\alpha \wedge \beta)$	$\sim (\alpha \wedge \beta)$	$\sim \alpha \vee \sim \beta$	$\sim (\alpha \wedge \beta) \Leftrightarrow \sim \alpha \vee \sim \beta$
F	F	T	T	F	T	T	T
F	T	T	F	F	T	T	T
T	F	F	T	F	T	T	T
T	T	F	F	T	F	F	T

The above truth table shows that $\sim (\alpha \wedge \beta) \Leftrightarrow \sim \alpha \vee \sim \beta$ is a tautology. We therefore conclude that $\sim (\alpha \wedge \beta) \equiv \sim \alpha \vee \sim \beta$. ■

Definition: Let $p_1, p_2, \ldots, p_k$ and q be wffs. $p_1, p_2, \ldots, p_k$ ***logically implies*** q, or q is a ***direct consequence*** of $p_1, p_2, \ldots, p_k$, denote by

$$\frac{p_1, p_2, \ldots, p_k}{q},$$

if $(p_1 \wedge p_2 \wedge \ldots \wedge p_k) \Rightarrow q$ is a tautology. We shall call

$$\frac{p_1, p_2, \ldots, p_k}{q}$$

an ***inference rule***.

A list of *inference rules* is provided in Table 1.6.1. The validity of these inference rules can be proven with the use of a truth table.

Example 1.6.2 Prove **I5**.

We have to prove that $((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$ is a tautology. This can be accomplished by using a truth table.

p	q	r	$(p \Rightarrow q)$	$(q \Rightarrow r)$	$(p \Rightarrow r)$	$((p \Rightarrow q) \wedge (q \Rightarrow r))$	$((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$
F	F	F	T	T	T	T	T
F	F	T	T	T	T	T	T
F	T	F	T	F	T	F	T
F	T	T	T	T	T	T	T
T	F	F	F	T	F	F	T
T	F	T	F	T	T	F	T
T	T	F	T	F	F	F	T
T	T	T	T	T	T	T	T

Since the column for $((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$ contains only the T value, the wff $((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$ is thus a tautology. Consequently, the claim that I5 is an inference rule stands. ■

1.7 Formal Proof

"A simple man believes every word he hears;
a clever man understands the need for proof."

Tautologies represent statement forms which are always true. For instance, the fact that $\sim (\omega \wedge \sim \omega)$ is a tautology indicates that statements such as "It is not true that diving is an aquatic sport and diving is not an aquatic sport" and "It is not true that either Newton discovered the Law of Motion or Einstein discovered the Theory of Relativity and neither Newton discovered the Law of Motion nor Einstein discovered the Theory of Relativity" are true statements. Therefore, tautologies are of prime importance in propositional logic. A major task in propositional logic is to develop methods for determining if a wff is a tautology. We have seen that the truth table is one such effective method. The use of the truth table relies on the meaning (semantics) of various components forming the wffs. This approach is called the *semantic approach*.

In this section, we shall develop another approach for identifying tautologies. This approach uses a predesignated set of formulas and rules to generate a sequence of wffs for reasoning and the reasoning is based solely on the form (syntax) and not the meaning (semantics) of wffs in the sequence. The approach is called the *syntactic appoach*.

Equivalences

In the following equivalences, α, β, γ are wffs in propositional logic.

E1.	$\alpha \wedge \sim \alpha \equiv false$	Law of Contradiction
E2.	$\alpha \vee \sim \alpha \equiv true$	Law of Excluded Middle
E3.	$\alpha \wedge \alpha \equiv \alpha$	
E4.	$\alpha \vee \alpha \equiv \alpha$	
E5.	$\alpha \wedge true \equiv \alpha$	
E6.	$\alpha \vee false \equiv \alpha$	
E7.	$\alpha \wedge false \equiv false$	
E8.	$\alpha \vee true \equiv true$	
E9.	$\alpha \wedge \beta \equiv \beta \wedge \alpha$	Commutative Law
E10.	$\alpha \vee \beta \equiv \beta \vee \alpha$	
E11.	$(\alpha \wedge \beta) \wedge \gamma \equiv \alpha \wedge (\beta \wedge \gamma)$	Associative Law
E12.	$(\alpha \vee \beta) \vee \gamma \equiv \alpha \vee (\beta \vee \gamma)$	
E13.	$\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$	
E14.	$\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$	
E15.	$\sim\sim \alpha \equiv \alpha$	
E16.	$\sim (\alpha \wedge \beta) \equiv (\sim \alpha \vee \sim \beta)$	DeMorgan's Law
E17.	$\sim (\alpha \vee \beta) \equiv (\sim \alpha \wedge \sim \beta)$	
E18.	$\alpha \Rightarrow \beta \equiv \sim \alpha \vee \beta$	
E19.	$\alpha \Rightarrow \beta \equiv \sim \beta \Rightarrow \sim \alpha$	
E20.	$(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha) \equiv \alpha \Leftrightarrow \beta$	

Inference Rules

In the following inference rules, p,q,r are wffs in propositional logic.

I1.	$\dfrac{p}{p \vee q}$	*addition*
I2.	$\dfrac{p \wedge q}{p}$	*simplification*
I3.	$\dfrac{p, p \Rightarrow q}{q}$	*modus ponens*
I4.	$\dfrac{\sim q, p \Rightarrow q}{\sim p}$	*modus tollens*
I5.	$\dfrac{p \Rightarrow q, q \Rightarrow r}{p \Rightarrow r}$	*hypotheical syllogism*
I6.	$\dfrac{p, q}{p \wedge q}$	*conjunction*

Table 1.6.1

Although the syntactic approach is an alternate method for deriving tautologies in propositional logic, it is the only method in other more powerful logics such as the First-order logic to be discussed in the next chapter. It is certainly a good idea to begin the study of this approach with propositional logic.

In the syntatic approach, tautologies are not derived from the truth table but from the equivalences and the inference rules. In these notes, the equivalences and inference rules we are allowed to use are those listed in Table 1.6.1. Note that as the variables in each equivalence and inference rule are not just propositional variables but wffs, each equivalence and inference rule actually represents infinitely many equivalences or inference rules.

Definition: Let α, β be wffs such that $\alpha \equiv \beta$ or $\beta \equiv \alpha$. Let S be a wff containing one or more occurrences of α (as sub-wffs). The wff obtained from S by replacing one or more occurrences of α with β is called a ***direct consequence*** of S by ***substitution***.

Example 1.7.3

1. Let S be $(p \vee q)$. By letting α be p in equivalence E15, we obtain $\sim\sim p \equiv p$. Replacing the p in $(p \vee q)$ with $\sim\sim p$, we obtain $(\sim\sim p \vee q)$. Hence, $(\sim\sim p \vee q)$ is a direct consequence of S by substitution.
2. Let S be $(\sim\sim p \vee q) \wedge (q \Rightarrow r)$. By letting α be $\sim p$ and β be q in equivalence E18, we obtain $(\sim\sim p \vee q) \equiv (\sim p \Rightarrow q)$. Replacing $(\sim\sim p \vee q)$ in $(\sim\sim p \vee q) \wedge (q \Rightarrow r)$ with $(\sim p \Rightarrow q)$, we obtain $(\sim p \Rightarrow q) \wedge (q \Rightarrow r)$. Hence, $(\sim p \Rightarrow q) \wedge (q \Rightarrow r)$ is a direct consequence of S by substitution.
3. Let S be $\sim\sim p \vee \sim\sim p$. By letting α be p in equivalence E15, we obtain $\sim\sim p \equiv p$. Replacing the second $\sim\sim p$ in $\sim\sim p \vee \sim\sim p$ with p, we obtain $\sim\sim p \vee p$. hence, $\sim\sim p \vee p$ is a direct consequence of S by substitution. ■

Definition: Let $\alpha_1, \alpha_2, \ldots, \alpha_k$ and β be wffs, and $\frac{\alpha_1, \alpha_2, \cdots, \alpha_k}{\beta}$ be an inference rule. The wff β is said to be a ***direct consequence*** of $\alpha_1, \alpha_2, \ldots, \alpha_k$ by virtue of the inference rule $\frac{\alpha_1, \alpha_2, \cdots, \alpha_k}{\beta}$.

Example 1.7.4

1. Let α_1 be $\sim (p \vee q)$ and α_2 be $(p \wedge q) \Rightarrow (p \vee q)$. By letting q be $(p \vee q)$ and p be $(p \wedge q)$ in I4, we obtain $\frac{\sim(p \vee q), (p \wedge q) \Rightarrow (p \vee q)}{\sim(p \wedge q)}$. Hence, $\sim (p \wedge q)$ is a direct consequence of $\sim (p \vee q)$ and $(p \wedge q) \Rightarrow (p \vee q)$ by virtue of the inference rule I4.

Definition: A ***derivation*** or ***proof*** of a wff α from a set of wffs Γ in propositional logic is a sequence of wffs, $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that $\alpha_n = \alpha$, and for each i, $1 \leq i \leq n$,

(i) α_i is a wff in Γ, or
(ii) α_i is of the form $\omega \vee \sim \omega$, called *axiom*, where ω is a wff, or
(iii) α_i is a direct consequence of some of the preceding wffs by substitution or by virtue of one of the inference rules.

Let $\Gamma = \{\gamma_1, \gamma_2, \cdots \gamma_k\}$. We shall write $\Gamma \vdash \alpha$ or $\gamma_1, \gamma_2, \cdots, \gamma_k \vdash \alpha$ to indicate that there is a proof of α from Γ.

Example 1.7.5 Let Γ be $\{p \Rightarrow (q \Rightarrow r), q\}$ and α be $p \Rightarrow r$. The following is a proof of $p \Rightarrow r$ from $\{p \Rightarrow (q \Rightarrow r), q\}$.

1.	$p \Rightarrow (q \Rightarrow r)$	from Γ
2.	$\sim p \vee (q \Rightarrow r)$	1,E18,substitution
3.	$\sim p \vee (\sim q \vee r)$	2,E18,substitution
4.	$(\sim p \vee \sim q) \vee r$	3,E12,substitution

5.	$(\sim q \vee \sim p) \vee r$	4,E10,substitution
6.	$\sim q \vee (\sim p \vee r)$	5,E12,substitution
7.	$q \Rightarrow (\sim p \vee r)$	6,E18,substitution
8.	q	from Γ
9.	$\sim p \vee r$	8,7,I3
10.	$p \Rightarrow r$	9,E18,substitution

Remark: When an inference rule, say $\frac{p_1, p_2, \cdots, p_k}{q}$, is used in a formal proof, each $p_i, 1 \leq i \leq k$, must appear as a wff, and *not* a sub-wff, in the sequence of wffs obtained thus far. The following are two illustrations.

(i) Let Γ be $\{\alpha \vee (\beta \Rightarrow \gamma), (\gamma \Rightarrow \delta)\}$ and ω be $\alpha \vee (\beta \Rightarrow \delta)$. The following is not a valid formal proof of ω from Γ.

1.	$\alpha \vee (\beta \Rightarrow \gamma)$	from Γ
2.	$(\gamma \Rightarrow \delta)$	from Γ
3.	$\alpha \vee (\beta \Rightarrow \delta)$	1,2, I5

The error occurs at Step 3, where inference rule I5 is applied to the sub-wff $(\beta \Rightarrow \gamma)$ in Step 1. The following is a valid proof:

1.	$\alpha \vee (\beta \Rightarrow \gamma)$	from Γ
2.	$(\gamma \Rightarrow \delta)$	from Γ
3.	$\alpha \vee (\sim \beta \vee \gamma)$	1, E18
4.	$(\alpha \vee \sim \beta) \vee \gamma$	3, E12
5.	$(\sim\sim \alpha \vee \sim \beta) \vee \gamma$	4,E15
6.	$\sim (\sim \alpha \wedge \beta) \vee \gamma$	5,E16
7.	$(\sim \alpha \wedge \beta) \Rightarrow \gamma$	6, E18
8.	$(\sim \alpha \wedge \beta) \Rightarrow \delta$	7,2, I5
9.	$\sim (\sim \alpha \wedge \beta) \vee \delta$	8, E18
10.	$(\sim\sim \alpha \vee \sim \beta) \vee \delta$	9, E16
11.	$(\alpha \vee \sim \beta) \vee \delta$	10, E15
12.	$\alpha \vee (\sim \beta \vee \delta)$	11, E12
13.	$\alpha \vee (\beta \Rightarrow \delta)$	12, E18

(ii) The following is not a valid proof of $\sim \alpha$ from $\sim (\alpha \wedge \beta)$.

1.	$\sim (\alpha \wedge \beta)$	from Γ
2.	$\sim \alpha$	1, I2

The error occurs in Step 2 where inference rule I2 is applied to the sub-wff $(\alpha \wedge \beta)$ in Step 1. As a matter of fact, there is no proof of $\sim \alpha$ from $\sim (\alpha \wedge \beta)$.

Definition A wff α is a ***theorem*** if $\Gamma \vdash \alpha$ where Γ is either empty or contains only theorems. A theorem whose sole purpose is to shorten the proof of another theorem is called a ***lemma***. A ***corollary*** is a theorem which is an immediate consequence of another theorem.

Example 1.7.6 Prove that $\vdash (p \Rightarrow (q \Rightarrow p))$.

1.	$(p \vee \sim p)$	Axiom

2. $(p \vee \sim p) \vee \sim q$ 1, I1
3. $(\sim p \vee p) \vee \sim q$ 2, E10
4. $\sim p \vee (p \vee \sim q)$ 3, E12
5. $\sim p \vee (\sim q \vee p)$ 4, E10
6. $\sim p \vee (q \Rightarrow p)$ 5, E18
7. $(p \Rightarrow (q \Rightarrow p))$ 6, E18

Hence, $\vdash (p \Rightarrow (q \Rightarrow p))$ ■

Proof Methods

There are a number of proof methods which have been widely used in proving theorems. Among them, the most well-known one is perhaps the *direct proof* method, whose validity is based on the following Deduction Theorem.

Deduction Theorem: Let Γ be a set of wffs and α, β be wffs such that $\Gamma, \alpha \vdash \beta$[1], then $\Gamma \vdash (\alpha \Rightarrow \beta)$.

Proof: Omitted. ■

When Γ is empty, we have: if $\alpha \vdash \beta$ then $\vdash (\alpha \Rightarrow \beta)$, which provides the basis for the direct proof method.

> ***Direct Proof***: To prove that $\vdash (\alpha \Rightarrow \beta)$, we may prove that $\alpha \vdash \beta$. i.e. we may derive a proof of β from α.

Example 1.7.7 Prove that $\vdash (p \Rightarrow q) \Rightarrow (p \Rightarrow (p \wedge q))$.

(Direct Proof)

1. $(p \Rightarrow q)$ hypothesis
2. $(\sim p \vee q)$ 1, E18
3. $(\sim p \vee q) \wedge true$ 2, E5
4. $true \wedge (\sim p \vee q)$ 3, E9
5. $(p \vee \sim p) \wedge (\sim p \vee q)$ 4, E2
6. $(\sim p \vee p) \wedge (\sim p \vee q)$ 5, E10
7. $\sim p \vee (p \wedge q)$ 6, E14
8. $p \Rightarrow (p \wedge q)$ 7, E18

Hence, $\vdash (p \Rightarrow q) \Rightarrow (p \Rightarrow (p \wedge q))$ ■

> ***Proof by Contradiction***: To prove that $\vdash \omega$, we may proceed to prove that $\sim \omega \vdash false$.

Justification: Suppose $\sim \omega \vdash false$. Then by the Deduction Theorem, we obtain $\vdash (\sim \omega \Rightarrow false)$. We can thus proceed as follows:

1. $(\sim \omega \Rightarrow false)$ $\vdash (\sim \omega \Rightarrow false)$
2. $(\sim\sim \omega \vee false)$ 1, E18,substitution
3. $(\omega \vee false)$ 2, E15,substitution
4. ω 3, E6,substitution

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[1] $\Gamma, \alpha \vdash \beta$ stands for $\alpha_1, \alpha_2, \cdots, \alpha_n, \alpha \vdash \beta$ where $\{\alpha_1, \alpha_2, \cdots, \alpha_n\} = \Gamma$.

Hence, $\vdash \omega$. ■

In particular, to prove that $\vdash (\alpha \Rightarrow \beta)$, we may proceed to prove that $(\alpha \wedge \sim \beta) \vdash false$

Justification: Let ω be $(\alpha \Rightarrow \beta)$. Then $\sim \omega$ becomes $(\alpha \wedge \sim \beta)$.

Example 1.7.8 Prove that:

$$\vdash (((p \vee \sim t) \Rightarrow (\sim u \vee q)) \wedge ((s \Rightarrow r) \Rightarrow p) \wedge ((\sim r \vee t) \Rightarrow \sim s)) \Rightarrow (u \Rightarrow q).$$

(Proof by Contradiction)

1.	$(((p \vee \sim t) \Rightarrow (\sim u \vee q)) \wedge ((s \Rightarrow r) \Rightarrow p) \wedge ((\sim r \vee t) \Rightarrow \sim s)) \wedge \sim (u \Rightarrow q)$	Hypothesis
2.	$((p \vee \sim t) \Rightarrow (\sim u \vee q)) \wedge ((s \Rightarrow r) \Rightarrow p) \wedge ((\sim r \vee t) \Rightarrow \sim s)$	1, I2
3.	$\sim (u \Rightarrow q) \wedge (((p \vee \sim t) \Rightarrow (\sim u \vee q)) \wedge ((s \Rightarrow r) \Rightarrow p) \wedge ((\sim r \vee t) \Rightarrow \sim s))$	1, E9
4.	$\sim (u \Rightarrow q)$	3, I2
5.	$((p \vee \sim t) \Rightarrow (\sim u \vee q)) \wedge (((s \Rightarrow r) \Rightarrow p) \wedge ((\sim r \vee t) \Rightarrow \sim s))$	2, E11
6.	$((p \vee \sim t) \Rightarrow (\sim u \vee q))$	5, I2
7.	$(((s \Rightarrow r) \Rightarrow p) \wedge ((\sim r \vee t) \Rightarrow \sim s)) \wedge ((p \vee \sim t) \Rightarrow (\sim u \vee q))$	5, E9
8.	$(((s \Rightarrow r) \Rightarrow p) \wedge ((\sim r \vee t) \Rightarrow \sim s))$	7, I2
9.	$((s \Rightarrow r) \Rightarrow p)$	8, I2
10.	$(((\sim r \vee t) \Rightarrow \sim s) \wedge ((s \Rightarrow r) \Rightarrow p))$	8, E9
11.	$((\sim r \vee t) \Rightarrow \sim s)$	10, I2
12.	$\sim (\sim u \vee q)$	4, E18
13.	$\sim (p \vee \sim t)$	6,12, I4
14.	$\sim p \wedge \sim\sim t$	13, E17
15.	$\sim p$	14, I2
16.	$\sim (s \Rightarrow r)$	9,15, I4
17.	$\sim (\sim s \vee r)$	16, E18
18.	$\sim\sim s \wedge \sim r$	17, E17
19.	$\sim\sim s$	18, I2
20.	$\sim (\sim r \vee t)$	11,19, I4
21.	$\sim\sim r \wedge \sim t$	20, E17
22.	$\sim t \wedge \sim\sim r$	21, E9
23.	$\sim t$	22, I2
24.	$\sim\sim t \wedge \sim p$	14, E9
25.	$\sim\sim t$	16, I2
26.	$\sim t \wedge \sim\sim t$	23,25, I6
27.	$false$	26, E1

Hence, $\vdash (((p \vee \sim t) \Rightarrow (\sim u \vee q)) \wedge ((s \Rightarrow r) \Rightarrow p) \wedge ((\sim r \vee t) \Rightarrow \sim s)) \Rightarrow (u \Rightarrow q)$. ■

The *Indirect proof* method reduces the task of deriving a proof for a theorem of the form $\vdash (\alpha \Rightarrow \beta)$ to that of $\vdash (\sim \beta \Rightarrow \sim \alpha)$.

> ***Indirect Proof***: To prove that $\vdash (\alpha \Rightarrow \beta)$, we may proceed to prove that $\vdash (\sim \beta \Rightarrow \sim \alpha)$.

Justification: Suppose $\vdash (\sim \beta \Rightarrow \sim \alpha)$ (i.e. $(\sim \beta \Rightarrow \sim \alpha)$ has been proven to be a theorem). Then,

1.	$(\sim \beta \Rightarrow \sim \alpha)$	$\vdash (\sim \beta \Rightarrow \sim \alpha)$
2.	$(\alpha \Rightarrow \beta)$	1,E19, substitution

Hence, $\vdash (\alpha \Rightarrow \beta)$. ■

Obviously, to prove that $\vdash (\sim \beta \Rightarrow \sim \alpha)$, we still have to use either the direct proof or proof by contradiction method.

Example 1.7.9 Prove that $\vdash (p \Rightarrow \sim p) \Rightarrow \sim p$.

(Indirect proof) We shall prove that $\vdash \sim\sim p \Rightarrow \sim (p \Rightarrow \sim p)$ using the direct proof method.

(Direct proof)

1.	$\sim\sim p$	assumption
2.	$\sim (\sim p \vee \sim p)$	1, E4
3.	$\sim (p \Rightarrow \sim p)$	2, E18

Hence, $\vdash \sim\sim p \Rightarrow \sim (p \Rightarrow \sim p)$.

Consequently, $\vdash (p \Rightarrow \sim p) \Rightarrow \sim p$. ■

The following two proof methods are not as general as the previous three methods. However, in situations where they can be applied, they have proven to be handy and lead to shorter proofs.

Vacuous Proof: To prove that $\vdash \alpha \Rightarrow \beta$, we may attempt to prove that $\vdash \alpha \Rightarrow false$.

Justification: Suppose $\vdash \alpha \Rightarrow false$. Then,

1.	$\alpha \Rightarrow false$	$\vdash \alpha \Rightarrow false$
2.	$\sim \alpha \vee false$	1, E18
3.	$\sim \alpha$	2, E6
4.	$\sim \alpha \vee \beta$	3, I1
5.	$\alpha \Rightarrow \beta$	4,E18,substitution

Hence, $\vdash \alpha \Rightarrow \beta$. ■

Example 1.7.10 Prove that $\vdash (p \wedge q) \wedge (p \Rightarrow \sim q) \Rightarrow (p \vee r)$.
(Vacuous proof)

1.	$(p \wedge q) \wedge (p \Rightarrow \sim q)$	hypothesis
2.	$\sim\sim (p \wedge q) \wedge (p \Rightarrow \sim q)$	1, E15
3.	$\sim (\sim p \vee \sim q) \wedge (p \Rightarrow \sim q)$	2, E16
4.	$\sim (p \Rightarrow \sim q) \wedge (p \Rightarrow \sim q)$	3, E18
5.	$(p \Rightarrow \sim q) \wedge \sim (p \Rightarrow \sim q)$	4, E9
6.	$false$	5, E1

Hence, $\vdash (p \wedge \sim p) \Rightarrow q$ ■

Trivial Proof: To prove that $\vdash \alpha \Rightarrow \beta$, we may attempt to prove that $\vdash \beta$.

Justification: Suppose $\vdash \beta$. Then,

1.	β	$\vdash \beta$
2.	$\beta \vee \sim \alpha$	1, I1
3.	$\sim \alpha \vee \beta$	2, E10
4.	$\alpha \Rightarrow \beta$	3, E18,substitution

Hence, $\vdash \alpha \Rightarrow \beta$. ■

Example 1.7.11 Example: Prove that $\vdash \alpha \Rightarrow (\beta \Rightarrow \beta)$.

(Trivial Proof)

1.	$(\beta \vee \sim \beta)$	axiom
2.	$(\sim \beta \vee \beta)$	1, E10
3.	$(\beta \Rightarrow \beta)$	2, E18

Hence, $\vdash \alpha \Rightarrow (\beta \Rightarrow \beta)$. ■

> ***Proof by Cases***: To prove that $\vdash (\alpha_1 \vee \alpha_2 \vee \cdots \vee \alpha_k) \Rightarrow \beta$, we may proceed to prove that $\vdash \alpha_1 \Rightarrow \beta$ and $\vdash \alpha_2 \Rightarrow \beta$ and . . . and $\vdash \alpha_k \Rightarrow \beta$.

Justification: Suppose $\vdash \alpha_1 \Rightarrow \beta$ and $\vdash \alpha_2 \Rightarrow \beta$ and . . . and $\vdash \alpha_k \Rightarrow \beta$. Then,

1.	$\alpha_1 \Rightarrow \beta$	$\vdash \alpha_1 \Rightarrow \beta$
2.	$\alpha_2 \Rightarrow \beta$	$\vdash \alpha_2 \Rightarrow \beta$
3.	. . .	. . .
4.	$\alpha_k \Rightarrow \beta$	$\vdash \alpha_k \Rightarrow \beta$
5.	$(\alpha_1 \Rightarrow \beta) \wedge (\alpha_2 \Rightarrow \beta) \wedge \cdots \wedge (\alpha_k \Rightarrow \beta)$	1,2, . . . ,4, I6 (repeated k-1 times)
6.	$(\sim \alpha_1 \vee \beta) \wedge (\sim \alpha_2 \vee \beta) \wedge \cdots \wedge (\sim \alpha_k \vee \beta)$	5, E18 (repeated k times)
7.	$(\beta \vee \sim \alpha_1) \wedge (\beta \vee \sim \alpha_2) \wedge \cdots \wedge (\beta \vee \sim \alpha_k)$	6, E10 (repeated k times)
8.	$\beta \vee (\sim \alpha_1 \wedge \sim \alpha_2 \wedge \cdots \wedge \sim \alpha_k)$	7, E14 (repeated k–1 times)
9.	$(\sim \alpha_1 \wedge \sim \alpha_2 \wedge \cdots \wedge \sim \alpha_k) \vee \beta$	8, E10
10.	$\sim (\alpha_1 \vee \alpha_2 \vee \cdots \vee \alpha_k) \vee \beta$	9, E17 (repeated k–1 times)
11.	$(\alpha_1 \vee \alpha_2 \vee \cdots \vee \alpha_k) \Rightarrow \beta$	10, E18

Hence, $\vdash (\alpha_1 \vee \alpha_2 \vee \cdots \vee \alpha_k) \Rightarrow \beta$. ■

Example 1.7.12 Prove that $\vdash ((p \wedge (p \Rightarrow \sim q)) \vee (q \Rightarrow \sim q)) \Rightarrow \sim q$.

(Proof by Cases)

Case 1: Prove that $\vdash (p \wedge (p \Rightarrow \sim q)) \Rightarrow \sim q$

(Direct proof)

1.	$p \wedge (p \Rightarrow \sim q)$	hypothesis
2.	p	1, I2
3.	$(p \Rightarrow \sim q) \wedge p$	1, E9
4.	$(p \Rightarrow \sim q)$	3, I2
5.	$\sim q$	2,4 I3

Hence, $\vdash (p \wedge (p \Rightarrow \sim q)) \Rightarrow \sim q$.

Case 2: Prove that $\vdash (q \Rightarrow \sim q) \Rightarrow \sim q$.

(Proof by Contradiction)

1.	$(q \Rightarrow \sim q) \wedge \sim\sim q$	Hypothesis
2.	$(q \Rightarrow \sim q)$	1, I2
3.	$\sim\sim q \wedge (q \Rightarrow \sim q)$	1, E9
4.	$\sim\sim q$	3, I2
5.	$\sim q$	4,2, I4

6.	$\sim q \wedge \sim\sim q$	5,4, I6
7.	$false$	6, E1

Hence, $\vdash (q \Rightarrow \sim q) \Rightarrow \sim q$.

Thus, $\vdash ((p \wedge (p \Rightarrow \sim q)) \vee (q \Rightarrow \sim q)) \Rightarrow \sim q$. ■

(**Remark:** compare the proof of Case 2 with that of Example 1.7.9)

The following special case of *proof by cases* is commonly used:

> To prove $\vdash \omega$, we may prove $\vdash \alpha_1 \Rightarrow \omega, \vdash \alpha_2 \Rightarrow \omega, \ldots, \vdash \alpha_k \Rightarrow \omega$ and $\vdash \sim(\alpha_1 \vee \alpha_2 \vee \cdots \vee \alpha_k) \Rightarrow \omega$.

Justification: Left as an exercise. ■

Example 1.7.13 Prove that $\vdash (p \Rightarrow \sim p) \Rightarrow \sim p$.

(Proof by Cases)

Case 1: Prove that $\vdash p \Rightarrow ((p \Rightarrow \sim p) \Rightarrow \sim p)$.

(Direct proof)

1.	p	hypothesis
2.	$\sim\sim p$	1, E15
3.	$\sim\sim p \wedge \sim\sim p$	2, E3
4.	$\sim(\sim p \vee \sim p)$	3, E17
5.	$\sim(p \Rightarrow \sim p)$	4, E18
6.	$\sim(p \Rightarrow \sim p) \vee \sim p$	5, I1
7.	$(p \Rightarrow \sim p) \Rightarrow \sim p$	6, E18

Therefore, $\vdash p \Rightarrow ((p \Rightarrow \sim p) \Rightarrow \sim p)$.

Case 2: Prove that $\vdash \sim p \Rightarrow ((p \Rightarrow \sim p) \Rightarrow \sim p)$.

(Direct proof)

1.	$\sim p$	hypothesis
2.	$\sim p \vee \sim(p \Rightarrow \sim p)$	1, I1
3.	$\sim(p \Rightarrow \sim p) \vee \sim p$	2, E10
4.	$(p \Rightarrow \sim p) \Rightarrow \sim p$	3, E18

Therefore, $\vdash \sim p \Rightarrow ((p \Rightarrow \sim p) \Rightarrow \sim p)$.

Hence, $\vdash ((p \Rightarrow \sim p) \Rightarrow \sim p)$. ■

(**Remark:** In the above example of proof by cases, $k = 1$, and α_1, ω are p, and $\vdash ((p \Rightarrow \sim p) \Rightarrow \sim p)$, respectively.)

> ***Bidirectional Proof***: To prove that $\vdash \alpha \Leftrightarrow \beta$, we may prove that $\vdash \alpha \Rightarrow \beta$ and $\vdash \beta \Rightarrow \alpha$.

Justification: Suppose $\vdash \alpha \Rightarrow \beta$ and $\vdash \beta \Rightarrow \alpha$. Then

1.	$\alpha \Rightarrow \beta$	$\vdash \alpha \Rightarrow \beta$
2.	$\beta \Rightarrow \alpha$	$\vdash \beta \Rightarrow \alpha$
3.	$(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)$	1,2,I6
4.	$(\alpha \Leftrightarrow \beta)$	3,E20